An exhaustive check of the sharp Hilbert-form conjecture $\Psi = H + 3|T| \le 3$ over all 2^{48} sign inputs at resolution 2

(our own conjecture, a finite case; not progress on C^*)

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Abstract

In an earlier preprint [1] we conjectured (Conjecture 7.1) that the functional $\Psi(F) = H(F) + 3|T(F)|$ of the dyadic triangular Hilbert form satisfies $\Psi \le 3\prod_v \|F_v\|_3$ for all real inputs, which would give $C^* = 3$. Here we check the *sign-input, resolution-2* case exhaustively. Over all 2^{48} sign inputs $F \in \{\pm 1\}^{48}$ at resolution $N = 2$ we find $\Psi \le 3$, with maximum exactly 3 (in the integer units of 1/64 used throughout: $S = 192$), and 0 violations. The enumeration runs over 2^{30} reduced pairs (F_0, F_1); 4 608 of them attain 3. (The independent auditor found, as a by-product, that 36 864 of the 2^{48} inputs attain $\Psi = 3$.) Three enumerations agree: two producer implementations (a Python float32 kernel under an exactness guard, and an integer-only C program) that *share* two reductions (a sign fix and an ℓ_1 reduction over F_2), and the auditor's own enumeration, which uses neither reduction (an exact diagonal sign gauge, with a literal maximum over all 2^{16} choices of F_2) and reproduces the producer's histogram bin by bin, times 512, in all five modes. Controls: the earlier $\max R = 2$ is reproduced; the Haar-free form reaches $N + 1 = 3$ for R and 5 for Ψ (7 356 928 pairs above 3); a planted weight 4 in place of 3 produces 741 888 violating pairs. **This is our own conjecture; an exhaustive check at $N = 2$ for sign inputs (two producer enumerations plus an independent auditor enumeration); not progress on C^*; Tier [C].**

What this is NOT. It says nothing about real-valued inputs, nothing about resolutions $N \ge 3$, and nothing about the value of C^*. There is no Lean or other proof object for the 2^{48} statement. The compiler and hardware are trusted, not checked (all runs were on one machine). No erratum for the earlier preprint is needed.

Markers. [L]: a cited source, taken as stated (here only our own earlier preprint). [C]: executed code with an argument written down but not machine-checked. [X]: a recorded absence, supporting nothing by itself. There is no Tier A (no Lean) in this paper. Identifiers `DHP-·-nnnn` refer to `ledger/ledger.json`.

1 Setting and question

Let Λ be the dyadic triangular Hilbert form of [2] and C^* the best constant in $\Lambda(F) \le C^* \prod_v \|F_v\|_{L^3(\mathbb{R}^2_+)}$. Our earlier preprint [1] proves $C^* \ge 3$ and conjectures $C^* = 3$ through the sharper statement (Conjecture 7.1, label `conj:psi` there):

$$H(F) + 3|T(F)| \;\le\; 3\prod_v \|F_v\|_3 \qquad \text{for all real step inputs.}$$

Write $\Psi(F) = H(F) + 3|T(F)|$. For *sign* inputs ($F_v[i,j] \in \{\pm 1\}$, every $\|F_v\|_3 = 1$), the claim is $\Psi(F) \le 3$.

Definitions (Tier [L], from [1]; `DHP-L-0001`). At resolution N let $n = 2^N$ and F_v be the $n \times n$ matrix of the step values on $[0,1)^2$. The top term is $T(F) = n^{-3} \sum_{i,j,k} F_0[i,j]F_1[j,k]F_2[k,i]$. The remaining part is $H(F) = \sum_{l=0}^{N-1} \sum_{I \in A_{-l}} |L_I(F)|$, where at level l there are 2^l dyadic intervals of $b = n/2^l$ cells and the triple $(p, q, p \oplus q)$ contributes (eq. (trace) of [1])

$$L_I = 2^l n^{-3} \operatorname{tr}\big(SF_0^{pq}\, SF_1^{q,p\oplus q}\, SF_2^{p\oplus q,p}\big), \qquad S = \text{diag(Haar signs)}.$$

The preprint proves $\Lambda = |T| + H$, $C_1 = 1$ at resolution 1 and, at resolution N, $|T| + |L_{000}| \le 1$ and $\sum_{\text{level } l} |L_I| \le 1$ per level (Proposition `prop:level`), so $\Psi \le 2 + 2|T| \le 4$ at $N = 2$; it computed $\max R = 2$ at $N = 2$ (night lane of 2026-10-07).

The question. Closing the gap from 4 to 3 at $N = 2$ for sign inputs is a finite problem with 2^{48} inputs ($F_0, F_1, F_2 \in \{\pm 1\}^{4\times 4}$). The preregistration (`data/preregistration.json`, sha256 `140a28ad...ae9be`, item DH-Psi2, lane W1-d) asked for an exhaustive check at $N = 2$, with a kill criterion of 60 core-hours projected after a 1% sample. The answer is that $\Psi \le 3$ holds.

2 Method

Integer form at $N = 2$ ($n = 4$), in units of $1/64$ (`DHP-C-0001`).

$$S_w(F) = w\,|T| + |L_0| + 2 \sum_{(p,q)\in\{(0,0),(0,1),(1,0),(1,1)\}} |L_{1,pq}|,$$

with $T = \sum_{ijk} F_0[i,j]F_1[j,k]F_2[k,i]$ (an integer, no normalisation), level 0 (one interval, triple $(0,0,0)$) carrying the Haar sign $h = +1$ on cells $0,1$ and -1 on cells $2,3$, and level 1 (two intervals of two cells, prefactor 2, the four admissible triples $(p,q,p\oplus q)$ having XOR 0) carrying $h_1 = +1$ on the even cell and -1 on the odd cell. Scales $k \ge 1$ sum to $|T|$ and scales $k \le -2$ vanish. Then $\Psi \le 3 \iff S_3 \le 192$. Five *modes* are run: `PSI` ($w = 3$), `R` ($w = 1$), `PLANT4` ($w = 4$, a planted wrong weight), `HF_R` and `HF_PSI` (the same with all Haar signs replaced by $+1$). The orientation of h_1 is immaterial: swapping the two cells inside one half interval in one coordinate flips the sign of every affected L_I and leaves $|L_I|$ unchanged (200/200 random inputs unchanged).

Two reductions shared by both producer enumerations (`DHP-C-0002`). Both are exact identities; AUDIT.md section 3 proves them.

1. *Sign fix.* T and every L_I are trilinear in (F_0, F_1, F_2), so $F_0 \to -F_0$ and $F_1 \to -F_1$ negate every term and preserve every |term| and every maximum over F_2. Fixing $F_0[0,0] = F_1[0,0] = +1$ loses nothing and leaves $2^{15} \cdot 2^{15} = 2^{30}$ pairs.
2. *ℓ_1 reduction over F_2.* For fixed (F_0, F_1) each of the six terms (T, the level-0 triple, the four level-1 triples) is a linear form $\langle c_t, F_2\rangle$ with positive weights absorbed in c_t. Using $|a| = \max_{s=\pm 1} sa$ and swapping the two finite maxima,

$$\max_{F_2\in\{\pm1\}^{16}} \sum_t |\langle c_t, F_2\rangle| = \max_{s\in\{\pm1\}^6} \max_{F_2} \Big\langle \sum_t s_t c_t, F_2 \Big\rangle = \max_{s\in\{\pm1\}^6,\, s_0=+1} \Big\| \sum_t s_t c_t \Big\|_1,$$

   since $s \to -s$ leaves the norm unchanged.

Enumeration 1: Python (`data/code/psi_exhaust.py`). A parameterised re-implementation of the night-lane kernel (the pinned file `h5_multiscale.py` was not edited). int64 coefficients, then a float32 matrix product that is exact because a guard checks every partial sum is below 2^{24}. 160 chunks (32 per mode), contiguous coverage of $F_0 \in [0, 32768)$. Every per-F_0 argmax (32 768 per mode) was rechecked by an independent block-slicing evaluator (`direct_value`) and by the independently authored `h5_exact.py`; all agree in every mode.

Enumeration 2: C (`data/code/psi_exhaust.c`). Written from eq. (trace) with no floating point; Gray-code order over the 32 sign patterns; the argmax recomputed by a literal triple sum.

Enumeration 3: the auditor's (`data/audit/gauge_enum.c`). It shares neither reduction. The map $(F_0, F_1, F_2) \mapsto (D_0F_0D_1,\ D_1F_1D_2,\ D_2F_2D_0)$ with D_v diagonal ± 1 leaves every term $\sum F_0[i,j]F_1[j,k]F_2[k,i]\,w(i,j,k)$ invariant for any weight w, because $d_i^2 e_j^2 g_k^2 = 1$. On (F_0, F_1) the induced action has order 2^{11}, kernel $\pm(I, I, I)$, and is free. Canonical representatives (F_0 with row 0 and column 0 all $+1$: 2^9 choices; F_1 with row 0 all $+1$: 2^{12} choices) give 2^{21} pairs, one per orbit. The producer's set $\{F_0[0,0] = F_1[0,0] = +1\}$ meets each orbit in exactly $2^9 = 512$ pairs, so the prediction is that the producer's histogram equals $512\times$ the auditor's. For each representative the program takes a *literal* maximum over all 2^{16} choices of F_2 (Gray-code order, six integer accumulators, no float) and counts the maximisers. About 0.8 core-hours for all five modes.

3 Results

mode	max S	max value	pairs > 192	pairs at max	auditor: at max $\times 512$
`PSI` ($w = 3$)	192	3	0	4 608	9×512
`R` ($w = 1$)	128	2	0	4 096	8×512
`PLANT4` ($w = 4$)	256	4	741 888	512	1×512
`HF_R` (Haar-free)	192	3	0	512	1×512
`HF_PSI` (Haar-free)	320	5	7 356 928	512	1×512

Each row is over all 2^{30} reduced pairs. The Python and C histograms of $\max_{F_2} S$ over the 2^{30} pairs are identical in every mode (`summary.json`: `hist_identical`, `max_identical`, `violations_identical` all true), and $512\times$ the auditor's histogram equals both, bin by bin, in all five modes (`data/audit/rerun_compare_hashcheck_output.txt`; `DHP-C-0003`, `DHP-C-0004`).

The `PSI` histogram. The maximum over F_2 of S_3 takes only the values below over the 2^{30} pairs (sum of counts $1\,073\,741\,824 = 2^{30}$).

S	64	72	80	88	96	104	112	120
count	231 936	1 179 648	13 803 520	50 659 328	137 048 064	234 127 360	301 187 072	200 409 088
S	128	136	144	152	160	168	192	
count	91 788 288	30 867 456	10 264 576	1 638 400	368 640	163 840	4 608	

The largest value below the maximum is 168 ($\Psi = 21/8$): every one of the $2^{30} - 4\,608$ reduced pairs that does not attain 3 has $\max_{F_2} \Psi \le 21/8$. (The histogram is of the per-pair maximum over F_2; it does not give the distribution of Ψ over all inputs.)

The maximisers. The constant input is among them ($T = 1$, $H = 0$, $\Psi = 3$) and so is the seed g of [1] ($H = \frac{3}{2}$, $T = \frac{1}{2}$, $R = 2$, $\Psi = 3$). The auditor's enumeration finds 9 canonical pairs at the maximum with 18 maximising F_2 summed over them; hence exactly $9 \times 2 \times 2^{11} = 36\,864$ of the 2^{48} sign inputs attain $\Psi = 3$ (`DHP-C-0007`; a fact not claimed by the producer).

Reduction-free slices. For F_0 codes $0, 1, 7000, 12345, 21845, 30000, 32767$ in the producer's encoding, the auditor took all 2^{15} values of F_1 with $F_1[0,0] = +1$ and a literal maximum over all 2^{16} values of F_2, without the ℓ_1 reduction (the $F_1[0,0] = +1$ fix is still applied): the per-slice histograms equal the producer binary's in 35 of 35 (code, mode) cases, including non-constant F_0 with `PLANT4` violations (code 1: 344 pairs above 192). This is reported in AUDIT.md section 2; the slice output was not saved as a separate file.

Rebuild and rerun. The auditor rebuilt `psi_exhaust.c` with `gcc -O3 -march=native` and reran both C halves: every output file is byte-identical to `data/c_enum/` and matches the recorded sha256, with $\max S = 192$ and 0 violating pairs for `PSI`. The Python float32 kernel was *not* rerun by the auditor; it is cross-checked through its histograms.

4 Controls

All run before the PSI numbers were counted (DHP-C-0005).

control	expected	obtained
positive: exhaustive max R	2	2 (4 096 maximising pairs)
positive: Ψ(constant)	3	3
positive: seed g	$H = \frac{3}{2}, T = \frac{1}{2}, R = 2, \Psi = 3$	the same (Python evaluators; the C code does not evaluate g)
negative: Haar-free max R	$N + 1 = 3$	3 (the bound of Proposition prop:level is attained)
negative: Haar-free Ψ exceeds 3	> 3	max 5; 7 356 928 pairs > 3
negative: planted $w = 4$ finds violations	yes	max 4 at the constant; 741 888 pairs > 3 in 7 360 values of F_0, including values strictly between 3 and 4 ($49/16 \ldots 7/2$)
evaluator agreement on random inputs	all agree	2000/2000 ($w = 1, 3, 4$), producer; 3000/3000 auditor

The planted control shows the search finds violations away from the constant input; a search that silently under-maximised would have been caught. The auditor's mutation test adds power on the definitions side: replacing the level-1 weight by 1, using the odd XOR class, or using the coarse Haar function at level 1 makes its evaluator disagree with direct_value on 181, 155 and 148 of 200 random inputs respectively, so the 3000/3000 agreement is not vacuous. Other auditor checks: $N = 1$ exhaustively (2^{12} inputs) gives $\max R = 1$ and $\max \Psi = 3$ (the preprint's theorem); gauge invariance of T and H on 200/200 random gauges.

Kill criterion and cost (DHP-C-0006). A 1% sample ($F_0 \in [0, 328)$, PSI) took 40.89 s, projecting $40.89 \times 32768/328/3600 = 1.135$ core-hours (1.136 in the file, a rounding difference) for PSI and about 5.7 for all five Python modes, against the preregistered 60: the criterion did not fire. Actual cost: Python 25 964 s summed over 160 chunks (cores 4 and 5, shared with the C run), C about 80 minutes wall on the same two cores; about 10 core-hours or less in total. The producer's session was cut off by a spend limit after the C enumeration (2026-10-28 18:29 UTC) and aggregation resumed on 2026-10-31; chunk writes are atomic and every mode has contiguous coverage. The preregistration hash is unchanged at start and end.

Elementary-attack pass. Proposition prop:level gives $\Psi \leq 2 + 2|T| \leq 4$ at $N = 2$. Closing the gap from 4 to 3 needs the even/odd coupling between T and the fine level, which does not tensorise. No short proof was found, and none is claimed.

5 Tiers

- [L]: the definitions and Conjecture 7.1, from our own preprint [1], taken as stated (DHP-L-0001).
- [C]: the headline (DHP-C-0008) rests on the exhaustive enumerations, the proofs of the two reductions and the audit; each enumeration is executed code, none is machine-checked. The first producer enumeration uses float32 that is exact under the stated 2^{24} guard; the second is pure-integer C; the auditor's is pure-integer C with a different reduction.
- Tier [A]: nothing. Lean was not called for, since there is no violation, and a 2^{48} enumeration is not a decide target.
- [X]: absences recorded in DHP-X-0001.

6 Audit corrections applied

The audit (AUDIT.md; a separate AI agent instance; producer $\neq$ verifier) accepts the mathematical claim and corrects wording and bookkeeping only; no correction changes a number. All six are applied here (the producer's data/RESULT.md and data/result.json are kept verbatim, as pre-audit records whose hashes are recorded, and this text supersedes them):

1. *Which conjecture.* The producer's note said "Conjecture 1"; the paper numbers it **Conjecture 7.1** (`conj:psi`). We state: the sign-input, resolution-2 case of Conjecture 7.1 of [1] holds (finite exhaustive check).
2. *Independence.* "Two independent full enumerations" is replaced by: two full enumerations that are independent implementations but *share* the $F_0[0,0] = F_1[0,0] = +1$ sign fix and the ℓ_1 reduction over F_2; a third enumeration, by the auditor, uses neither and reproduces 512× its histogram bin by bin in all five modes (Section 3).
3. *Tier wording.* "Two independent implementations agreeing on full histograms" becomes: two producer implementations plus an auditor enumeration with a different reduction, agreeing on full histograms.
4. *Untracked binary.* `result.json` lists `code/psi_exhaust`, a machine-specific `-march=native` build artifact not tracked by git; a fresh clone would report it missing. It is not shipped here. Of the 199 entries of `files_sha256`, 198 match the shipped copies (script files travel as `.sh.txt`); the one that cannot be checked is that binary (`ledger/hash_check_paper_copy.txt`). The auditor checked all 199 against the lane directory.
5. *Residual common-mode risk.* The producer's sentence on a residual common-mode risk is replaced by: both shared reductions are exact identities (proved in AUDIT.md section 3) and were confirmed by an enumeration that uses neither. The parity convention of h_1 is immaterial (Section 2).
6. *Wording kept everywhere. Our own conjecture; exhaustive check at $N = 2$ for sign inputs (two producer enumerations plus an independent auditor enumeration); not progress on C^*; Tier [C].* The auditor's optional new fact, that 36 864 of the 2^{48} sign inputs attain $\Psi = 3$, is included.

7 Claim ledger

`ledger/build_ledger.py` files 10 claims: 1 at Tier [L] (`DHP-L-0001`), 8 at [C] (`DHP-C-0001..0008`, the last being the headline) and 1 at [X] (`DHP-X-0001`, recording what was not done). Each carries a content-addressed evidence blob in `ledger/evidence/`. There is no Tier [A] row (no Lean), so the flag `LEDGER_UNAUDITED_TIER_A` cannot arise. The gate (`tools/ledger.py ledger.json --evidence-dir evidence`; verbatim in `ledger/gate_output.txt`) reports 10 claims checked, no findings, exit code `0`. That says the bookkeeping holds; it licenses nothing about truth.

8 Limitations

- **Scope.** Sign inputs, resolution $N = 2$ only. Nothing for real-valued inputs, for $N \geq 3$ (where $2^{3 \cdot 4^N}$ makes the same method infeasible), or for C^*. A violation at higher resolution, or for non-sign inputs, is not excluded. This is a finite verification of our own conjecture in a special case, nothing more.
- **No proof object.** No Lean or kernel certificate for the 2^{48} statement; Tier [C].
- **AI audit only.** The audit is by a different AI agent instance; there is no human audit (owed).
- **Trusted base.** The compiler and the hardware are trusted, not checked. All three enumerations ran on the same machine: two through gcc, one through numpy/BLAS.
- **Not rerun.** The Python float32 kernel was not rerun by the auditor (only compared through histograms). The reduction-free slices (35/35) are reported in AUDIT.md and were not saved as a file. The auditor's scratch directory was `/tmp`; the shipped copies are in `data/audit/` and refer to `/tmp/dhaudit` paths inside the scripts.
- **Deviations from the preregistration.** (1) The preregistration names the `tensor/` evaluator; the exhaustive F_2-reduction kernel lives in `night/h5_multiscale.py`, which was not modified (a parameterised copy is `psi_exhaust.py`, and `h5_exact.py` was imported read-only). (2) An independent integer-only C enumeration and rechecks of every per-F_0 argmax were added. (3) Lane W1-d was budgeted for one core; cores 4 and 5 were used, as assigned.

- **Novelty.** The conjecture is ours [1]; to our knowledge it has not been addressed elsewhere. No novelty search was preregistered ("none needed").
- Not a physical claim of any kind.

9 Reproducibility

Repository AutoevolveAI, branch `worktree-openai-math-discovery`; HEAD when this paper was written: `1e29c538f2ceacd0e644fc990cfba32ce45fa0d7`; the DH-Psi2 outputs and this paper are uncommitted working-tree files at that time. Inputs are in `data/`: `code/` (`psi_exhaust.c` and `.c.txt`, `psi_exhaust.py`, drivers; the compiled binary is not shipped), `chunks/` (160), `c_enum/`, `summary.json`, `controls.json`, `logs/`, `audit/` (the auditor's `gauge_enum.c`, `aud_eval.py`, its histograms), `RESULT.md` and `result.json` (the producer's pre-audit text), `preregistration.json`. The command lines are summarised from the code headers.

```
gcc -O3 -march=native -o psi_exhaust code/psi_exhaust.c        # usage: psi_exhaust WT HAAR FO_LO FO_HI [bruteforce_pairs]
./psi_exhaust 3 1 0 16384 ; ./psi_exhaust 3 1 16384 32768       # PSI mode (WT=3, Haar) in two halves; code/c_runs.sh.txt runs
      all five modes
gcc -O3 -march=native -o gauge_enum audit/gauge_enum.c ; ./gauge_enum 1 ; ./gauge_enum 0   # auditor: Haar / Haar-free
python -I audit/compare.py                                      # 512 x auditor histogram vs producer histograms
python -I ledger/build_ledger.py ; python -I ledger/run_gate.py
```

References

[1] X. Callens, *The sharp constant of the absolute dyadic triangular Hilbert sum is at least 3*, preprint v1, 2026-10-08, Zenodo doi:10.5281/zenodo.23232389 (our own work; Conjecture 7.1, Proposition on levels, eq. (trace)) ([L]).

[2] OpenAI, *An L^3 bound for the dyadic triangular Hilbert form*, preprint, 5 October 2026, `github.com/openai/math`, as cited in [1]: the bound $\Lambda \le 40 \prod_v \|F_v\|_3$.